

Sample Project Documentation

Date	30 September 2026	Team ID	6ab22029e880316a482c5f2a
Project Name	EduGenie – AI-Powered Educational Assistant	GitHub	https://github.com/theopintheo/EduGenie
Team Members	Theopin B, Antony Ragul A, Saravanan S, Karthick M	Status	Completed / working implementation

1. Executive Summary

EduGenie is a modular browser-based AI educational assistant for self-learners. It combines question answering, beginner-friendly explanations, summarization, quiz generation and learning-path recommendations in one lightweight interface.

2. Problem

Learners often switch between search engines, videos, notes, quiz sites and planning tools. This fragmentation increases study friction. EduGenie brings common learning tasks into one structured workflow.

3. Target Users

College students, school learners, beginners and self-learners needing quick explanations, revision summaries, practice questions or a route through a new topic.

4. Core Modules

Module	Responsibility
main.py	FastAPI app, routes, templates/static mounting, validation and error handling.
ai_service.py	Model client, configuration, generation and demo fallback.
qna.py	Question-answering prompt.
explanation_module.py	Beginner explanation with analogy and takeaways.
summary_module.py	Plain-language summaries.
quiz_module.py	Structured quiz generation and validation.
learning_path.py	Beginner-to-advanced recommendations.
templates/index.html	Single-page workspace.
static/app.js	Tool switching, API calls, rendering and quiz checking.
static/style.css	Responsive visual design.

5. API Surface

Endpoint	Method	Purpose
/health	GET	Application and AI status.
/qa	POST	Question answering.
/explain	POST	Topic explanation.
/summarize	POST	Study-note summarization.
/quiz	POST	Quiz generation.
/learn/recommendations	POST	Learning-path recommendations.
/docs	GET	Interactive OpenAPI docs.

6. Responsible AI and Security

API credentials should remain in environment variables. Contributors should never commit keys. Generated educational content should be reviewed against trusted sources. Public deployment should add authentication, rate limiting and observability.

7. Current Limitations

The supplied implementation does not include persistent accounts/history, database-backed progress, production authentication, rate limiting, multilingual output, formal automated test coverage, verified production deployment or measured load-test results.

8. Future Roadmap

Phase	Enhancement	Impact
2	Authentication, study history, progress dashboard	Personalized continuity
3	Multilingual output, PDF/image input, accessibility upgrades	Broader usability
4	Adaptive learning, teacher/parent dashboards, group study	Advanced personalization

9. Team

Member	Contribution
Theopin B	AI/model architecture, integration, functional testing, final review
Antony Ragul A	Frontend UI, responsive styling, quiz interaction, UX review
Saravanan S	FastAPI backend, REST APIs, validation, API documentation
Karthick M	Environment setup, run/deployment verification, smoke testing

10. Conclusion

EduGenie demonstrates a complete modular generative-AI learning workflow with clear separation between UI, API routes, feature prompts and the shared AI provider service. The architecture provides a practical foundation for personalization, persistence, accessibility and deployment hardening.